

DEQ SITE IDENTIFICATION FORM

Complete the following information and provide the attachments listed below for each new site or existing site that has not had a Site Identification Form with attachments submitted to the DEQ since January 1, 1998. Please amend any previous Site Identification Forms when it becomes known that any of the information requested below has changed (i.e. new land owner).

FACILITY NAME City of Cadillac WWTB		NPDES or COC PERMIT NUMBER MI0020257	
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SITE IDENTIFICATION FORM				
A. Site ID	Site ID # 21N09W25-AB01	Street or Route # West of Seeley Rd, North of 50 Rd		
B. County/Township:	County Wexford	Township Clam Lake		
C. State Planar Coordinates:	SE 1/4	SE 1/4	Section 25	Town T21N
D. Lat/Long (nearest second):	Latitude 44° 10' 53.93"N		Longitude 85° 20' 14.82"W	Method Used (i.e. GPS, topo) Google Earth
E. Site Ownership	Name of Owner Alan Benson		Phone 231-920-6783	
Street or P.O. Box 11799 East 50 Mile Rd				
City or Town Cadillac			State MI	Zip 49601
Is there a written agreement from the owner to land apply? ☒ Yes ☐ No, written agreement is required			Previous landowners name Hartley Benson	
			Property purchase date N/A	
Name of Lessee (if different)			Phone	
Street or P.O. Box				
City or Town			State	Zip
F. Site Characteristics	Site Type ☒ Agricultural ☐ Forest ☐ * Reclamation Site ☐ ** Lawn/Home Garden *Separate approval is required **Requires EQ Biosolids		Soils Information Date of last soils analysis <u>11-15-2019</u> *Phosphorous <u>76</u> units <u>PPM</u> Potassium <u>144</u> units <u>PPM</u> pH <u>7.0</u> Soil type(s) <u>Sandy</u> *300 lb/ac Bray P1 maximum	
Expected crops/vegetation to be grown				
Crop/Vegetation <u>Corn</u>		Nitrogen Requirement <u>135</u> lbs. N/acre _____ lbs. N/acre _____ lbs. N/acre		
Total acreage of site <u>46.2</u>		Percent slope Highest percent slope of the site? ☐ 0-6% ☐ 6-12% ☒ *Higher (Requires an approved site reclamation plan)		
Acreage used for crops <u>46.2</u>		If agricultural, is the site tilled? ☐ Yes ☐ No ☒ NA		
Public Exposure Is this land with a high potential for public exposure? (See Definitions section on Page 19) ☐ Yes ☒ No				

On an attached sheet provide a copy of the following:

- A Farm Services Agency map (formerly ASCS) of the proposed application site with property boundaries, surface waters, and discharge location of field tiles when present.
- A copy of the soil test analysis that is no older than 2 years old at the time of land application.
- A copy of the plat map (or other map showing property dimensions) with the proposed land application site highlighted.
- A soil survey map with the land application site identified.



CITY OF CADILLAC WASTEWATER TREATMENT PLANT

BIOSOLIDS RECYCLING PROGRAM PARTICIPATION

I hereby agree to participate in the City of Cadillac WWTP biosolids recycling program. I understand that biosolids will be applied to my land in accordance with federal EPA regulations and guidelines of the Michigan Department of Environmental Quality.

I understand that biosolids have been analyzed to be sure they can be applied to agricultural land in accordance with federal regulations and MDEQ guidelines. I understand that biosolids contain some heavy metals and that they must be below the EPA allowable limits for land application. I understand that the nutrients and/or the metal concentration may limit the frequency of biosolids application to my land.

My land is used for the production of agricultural crops. Information concerning crop and site restrictions has been explained to me. I agree to allow representatives of the MDEQ, Cadillac Waste Water Treatment Plant and the City's Injection contractor to have access to my land to obtain soil samples and for application of biosolids.

If I decide to cancel this agreement, I will give 60 days advance notice to the City of Cadillac WWTP.

Landowner: Alan Benson Signed: Alan Benson

Address: 11799 East 50th Rd Date: 10-29-19

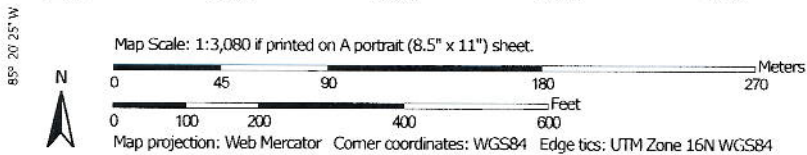
Township: Clam Lake Section: 25 Acres: 50

City WWTP Field Number: 21N09W25-AB01

If Rented, Farmer: _____

Address: _____

Soil Map—Lake and Wexford Counties, Michigan (21N09W25-AB01)



MAP LEGEND

	Area of Interest (AOI)		Spoil Area
	Area of Interest (AOI)		Stony Spot
	Soils		Very Stony Spot
	Soil Map Unit Polygons		Wet Spot
	Soil Map Unit Lines		Other
	Soil Map Unit Points		Special Line Features
	Special Point Features		Water Features
	Blowout		Streams and Canals
	Borrow Pit		Transportation
	Clay Spot		Rails
	Closed Depression		Interstate Highways
	Gravel Pit		US Routes
	Gravelly Spot		Major Roads
	Landfill		Local Roads
	Lava Flow		Background
	Marsh or swamp		Aerial Photography
	Mine or Quarry		
	Miscellaneous Water		
	Perennial Water		
	Rock Outcrop		
	Saline Spot		
	Sandy Spot		
	Severely Eroded Spot		
	Sinkhole		
	Slide or Slip		
	Sodic Spot		

MAP INFORMATION

The soil surveys that comprise your AOI were mapped at 1:15,800.

Warning: Soil Map may not be valid at this scale.

Enlargement of maps beyond the scale of mapping can cause misunderstanding of the detail of mapping and accuracy of soil line placement. The maps do not show the small areas of contrasting soils that could have been shown at a more detailed scale.

Please rely on the bar scale on each map sheet for map measurements.

Source of Map: Natural Resources Conservation Service
Web Soil Survey URL:
Coordinate System: Web Mercator (EPSG:3857)

Maps from the Web Soil Survey are based on the Web Mercator projection, which preserves direction and shape but distorts distance and area. A projection that preserves area, such as the Albers equal-area conic projection, should be used if more accurate calculations of distance or area are required.

This product is generated from the USDA-NRCS certified data as of the version date(s) listed below.

Soil Survey Area: Lake and Wexford Counties, Michigan
Survey Area Data: Version 13, Sep 16, 2019

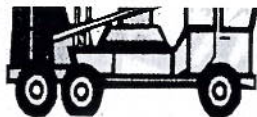
Soil map units are labeled (as space allows) for map scales 1:50,000 or larger.

Date(s) aerial images were photographed: May 9, 2014—Sep 19, 2016

The orthophoto or other base map on which the soil lines were compiled and digitized probably differs from the background imagery displayed on these maps. As a result, some minor shifting of map unit boundaries may be evident.

Map Unit Legend

Map Unit Symbol	Map Unit Name	Acres in AOI	Percent of AOI
12C	Emmet-Montcalm complex, 6 to 12 percent slopes	13.4	29.0%
12D	Emmet-Montcalm complex, 12 to 18 percent slopes	16.2	35.0%
12E	Emmet-Montcalm complex, 18 to 40 percent slopes	12.7	27.5%
20C	Montcalm-Graycalm complex, 6 to 12 percent slopes	0.0	0.1%
20E	Montcalm-Graycalm complex, 12 to 30 percent slopes	2.2	4.7%
22	Tawas-Roscommon association	1.3	2.9%
W	Water	0.3	0.7%
Totals for Area of Interest		46.2	100.0%



CLAM LAKE T21N • R9W

7000

08000

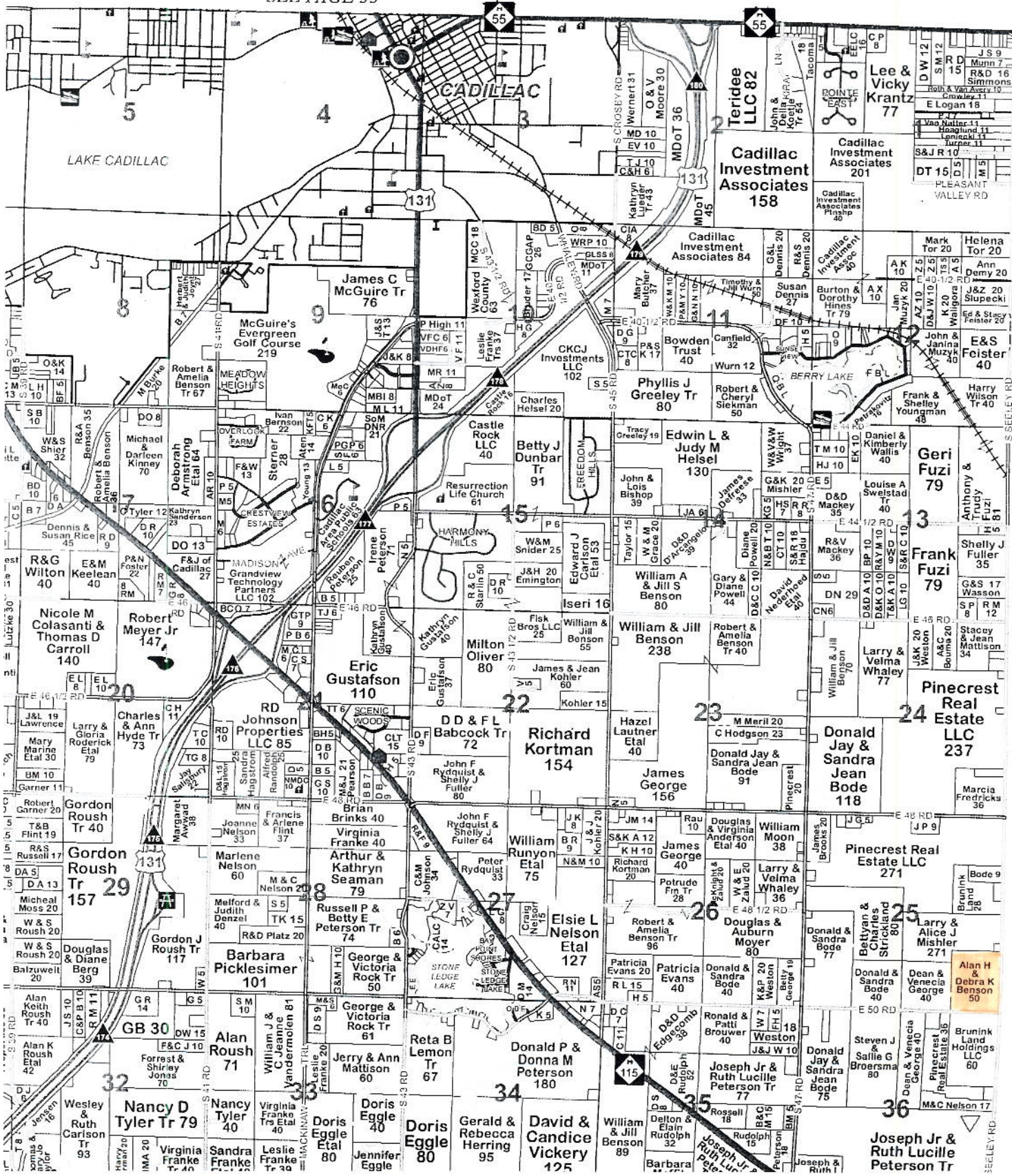
09000

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SEE PAGE 33



MISSAUKIE COUNTY

SEELEY RD

MICHIGAN STATE UNIVERSITY

MICHIGAN STATE UNIVERSITY
SOIL AND PLANT NUTRIENT LABORATORY
EAST LANSING, MICHIGAN 48824-1325
(517) 355-0218

SOIL TEST REPORT FOR:				CONSULTANT			
CITY OF CADILLAC 200 N. LAKE ST. CADILLAC MI 49601				WEXFORD COUNTY MSUE 401 N. LAKE STREET83 CADILLAC MI 49601 231-779-9480			
DATE	LAB #	COUNTY	Previous Crop	ACRES	FIELD ID	SOIL	TEXTURE
11/15/2019	240359	Wexford	Corn silage	50	21n09w25-ab01	Mineral	

SOIL NUTRIENT LEVELS			Below Optimum	Optimum	Above Optimum
¹ Soil pH	7.0	Lime Index			
² Phosphorus (P)	76	ppm	■■■■■■■■■■	■■■■■■■■■■	■■■■■■■■■■
³ Potassium (K)	144	ppm	■■■■■■■■■■	■■■■■■■■■■	■■■■■■■■■■
³ Magnesium (Mg)	89	ppm	■■■■■■■■■■	■■■■■■■■■■	■■■■■■■■■■

ADDITIONAL RESULTS:					Optional Tests:						
Calcium (Ca) (ppm)	CEC (meq/100 g)	% of Exchangeable Bases			Micronutrients (ppm)					Organic Matter %	Nitrate-N ppm
		K	Mg	Ca	B	Cu	Mn	Zn	Fe		
1228	7.3	5.1	10.2	84.7							

RECOMMENDATIONS:

Limestone: NONE

Tillage Depth: 9 inches

Target pH = 6.5

% Stand:

Plant Nutrients:					Micronutrient: (Optional)				
Year	Crop	Expected Yield	Nitrogen (lb N/A)	Phosphate (lb P ₂ O ₅ /A)	Potassium (lb K ₂ O/A)	Boron (lb B/A)	Manganese (lb Mn/A)	Zinc (lb Zn/A)	Copper (lb Cu/A)
1	Corn silage	18 ton	135	0	0	0.0			

Use a starter fertilizer high in nitrogen (30-40 lb N/A) when planting into heavy residue.

For questions about interpreting your soil results go to the following website: <http://msue.anr.msu.edu/experts> and contact the MSUE field crops agent in your area.